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Wheat disease recognition method based on the SC-ConvNeXt network model

Tianliang Dong[1,2], Xiao Ma[3], Bin Huang[2], Wenyu Zhong[2], Qingan Han[2], Qinghai Wu[1,2] & You Tang[1,2]✉

When utilizing convolutional neural networks for wheat disease identification, the training phase typically requires a substantial amount of labeled data. However, labeling data is both complex and costly. Additionally, the model's recognition performance is often disrupted by complex factors in natural environments. To address these issues, this paper proposes a wheat disease identification model, SC-ConvNeXt, which integrates the SimCLR pre-training framework and an improved CBAM attention mechanism. The model employs ConvNeXt-T as the feature extraction network. Initially, it uses the self-supervised SimCLR pre-training framework to learn inter-class similarities, reducing the reliance on labeled data during training. Subsequently, the CBAM attention module is integrated into ConvNeXt-T to enhance the model's feature extraction and generalization capabilities in complex backgrounds, and each attention module's loss function is improved with a LeakyReLU activation function to prevent neuron deactivation when inputs are negative. Furthermore, by introducing the Focal Loss function, the model addresses the imbalance in the quantity of easy and difficult-to-classify samples. The dataset used in this study comes from the 'Smart Agriculture' platform of Jilin Agricultural Science and Technology College, including images of three wheat diseases and healthy wheat. After expanding the dataset with various data augmentation methods, the effectiveness of adding SimCLR and the attention mechanism was sequentially verified. Comparative experiments were also conducted against four classic classification models. The experimental results show that the proposed SC-ConvNeXt model achieves an average classification accuracy of 88.05% on the test set, the highest among all comparative models. The model does not require additional labeled data during training, demonstrating its effectiveness in enhancing wheat disease recognition performance under natural environmental conditions without the need for extra labeled data training.

According to statistics, the global wheat production in the 2022/2023 fiscal year reached a new peak of 79 million tons, of which China's output accounted for approximately 18%, amounting to 135.76 million tons[1]. As the world's leading staple crop, wheat, through processing, can be made into bread, noodles, and even feed, playing a pivotal role in both human consumption and animal husbandry[2,3]. China, as the largest wheat-producing country globally, has long been plagued by wheat diseases, with domestic reductions in yield and quality due to these diseases being exceedingly common[4,5]. Wheat diseases can be categorized into several types, including fungal, bacterial, and viral diseases, among which the most common ones are wheat scab[6], powdery mildew[7], stripe rust[8] and so on.

To mitigate losses caused by wheat diseases, effective prevention and control are of paramount importance. However, the prevention and control measures differ significantly among various wheat diseases, making the timely and accurate identification of the type of wheat disease a key prerequisite for disease management. Currently, the primary methods for identifying wheat diseases still rely on manual recognition or expert diagnosis, which are inefficient, often delayed, and prone to misdiagnosis, failing to meet the demands of modern agricultural development[9,10]. Meanwhile, with the development and progress of artificial intelligence technology, intelligent identification methods for crop diseases have gradually become a new focus in agricultural production. Nowadays, many scholars have dedicated their research to the development of intelligent algorithms for this purpose[11,12].

[1]School of Information and Control Engineering, Jilin University of Chemical Technology, Jilin 132022, Jinlin, China. [2]School of Electrical and Information Engineering, Jilin Agricultural Science and Technology University, Jilin 132101, Jilin, China. [3]School of Computer Science and Information Engineering, Qilu Institute of Technology, Jinan 250200, Shandong, China. ✉email: tangyou@jlnku.edu.cn

Effective prevention and control are essential for minimizing losses from wheat diseases. However, the measures for preventing and controlling various wheat diseases differ significantly. Therefore, timely and accurate identification of wheat disease types is a crucial prerequisite for effective disease management. Currently, the primary methods for identifying wheat diseases still rely on manual inspection or expert diagnosis. These approaches face challenges, including inefficiency, delayed detection, and susceptibility to misdiagnosis, making it difficult to meet the demands of modern agricultural development[9,10]. Meanwhile, advancements in artificial intelligence have led to its deep and widespread application across fields like urban transportation and industrial engineering[11–13]. Increasingly, intelligent methods for identifying plant diseases have also become a prominent area of interest in agricultural production. Many researchers are now focused on developing intelligent algorithms for this purpose[14,15].

In early research, studies on crop disease recognition were mostly based on traditional machine learning methods, which often required manual feature design and extraction. This process was not only tedious but also demanded researchers have substantial related knowledge and extensive experience, limiting the universality and scope of application of these methods[16]. In recent years, thanks to advancements in computing power and the emergence of large-scale data labeling, Convolutional Neural Networks (CNN) have entered a golden era of development. Image recognition methods based on CNNs eliminate the cumbersome feature extraction step, opting for an end-to-end structure instead. This data-driven approach can automatically extract deep abstract features, avoiding the incompleteness that might arise from manual feature design[17]. As a result, an increasing number of scholars are combining convolutional neural networks with crop disease recognition. Lv et al.[18] proposed a new recognition method based on the AlexNet architecture, incorporating batch normalization techniques to address overfitting issues and using PReLU activation functions and the Adaband optimizer to enhance the model's convergence speed and accuracy. Li et al.[19] utilized a residual network structure to identify apple scab, black star disease, cedar rust, and healthy leaves. Compared to VGG and AlexNet, their network structure significantly improved recognition accuracy. Miao et al.[20] adopted an improved VGG network model to recognize diseases in cucumbers and rice on the small dataset AES-IMAGE, adjusting the pre-trained model to achieve this. Lin et al.[21] introduced GrapeNet, a lightweight model tailored to address the issue of morphological diversity in grape leaf diseases. This model can recognize grape leaf diseases at different symptomatic stages, successfully increasing recognition accuracy while reducing the number of model parameters.

Despite the successes of these algorithms in identifying diseases in some crops[18–21], existing research on intelligent recognition of wheat diseases remains relatively limited, and is mostly conducted against a single background. Currently, wheat disease images are primarily collected in complex field environments, coupled with the varying difficulty of classification among the data samples, which greatly limits the model's robustness and generalizability. Additionally, model training often involves a significant amount of data labeling, a process that is both complex and costly, further increasing the difficulty of the training process. To address these issues, this study proposes the SC-ConvNeXt model for wheat disease recognition based on the ConvNeXt-T network. The main findings of this study can be summarized as follows: (1)The ConvNeXt-T is used as the feature extraction network, combined with the SimCLR pre-training framework, to reduce the use of labeled data during the training phase through self-supervised training, thereby speeding up training; (2) To overcome the negative impacts of complex backgrounds in the images, the Convolutional Block Attention Module (CBAM) is integrated into the feature extraction network, enabling the model to focus on wheat disease features and enhance recognition efficiency; (3) The attention module is improved by using the LeakyReLU activation function instead of ReLU, to prevent neuron deactivation when inputs are negative; (4) Focal Loss is introduced to address the imbalance between easy and difficult-to-classify samples.

Materials and method

Dataset and data augmentation

The wheat dataset employed in this study was sourced from the "Smart Agriculture" platform of the Jilin Agricultural Science and Technology College, located in Jilin Province, China. The dataset comprises 2,028 images taken against the natural background of the wheat fields at the college. To mimic the real-world challenges of identifying wheat diseases under complex environmental conditions and to enhance the model's generalizability, robustness, and prevent overfitting, various data augmentation techniques were applied. These techniques included random brightness adjustments, Gaussian noise, occlusions, and rotations (ranging from -180 degrees to 180 degrees). The random brightness feature simulates various lighting conditions in the real world, such as sunny and cloudy days, by adjusting the brightness of the image randomly. The random Gaussian noise function blurs the image by adding noise to replicate conditions like rainy and dusty weather, or even low-quality images resulting from outdated capture equipment. The random occlusion feature emulates a variety of real-world occlusions. These include random blocking of parts of the image by overlapping leaves, natural elements like weeds, soil clods, or dew, as well as areas with insufficient brightness or shadow caused by sunlight, clouds, plants, and other factors. The random rotation process simulates the diverse angles and orientations of wheat in an actual agricultural setting by randomly rotating the image. This technique captures varied shooting angles, natural growth patterns, and the swaying of plants due to wind.

After augmentation, the dataset expanded to 10,140 images.This dataset includes images of healthy wheat as well as images depicting three common wheat diseases: leaf rust, wheat scab, and stripe rust. The images were divided in a ratio of 1:7:2 for different phases: self-supervised learning training, supervised learning training, and testing. Notably, images used in the self-supervised learning phase did not require class labels. The distribution of images across different disease categories is detailed in Table 1, and illustrations of the various wheat disease images are shown in Fig. 1. Images after augmentation are depicted in Fig. 2.

Name	Label	Original images	After data augmentation
Healthy	0	516	2580
Leaf Rust	1	520	2600
Wheat Scab	2	496	2480
Wheat sharp eyespot	3	496	2480
Total	-	2028	10,140

Table 1. Specific distribution of disease quantity.

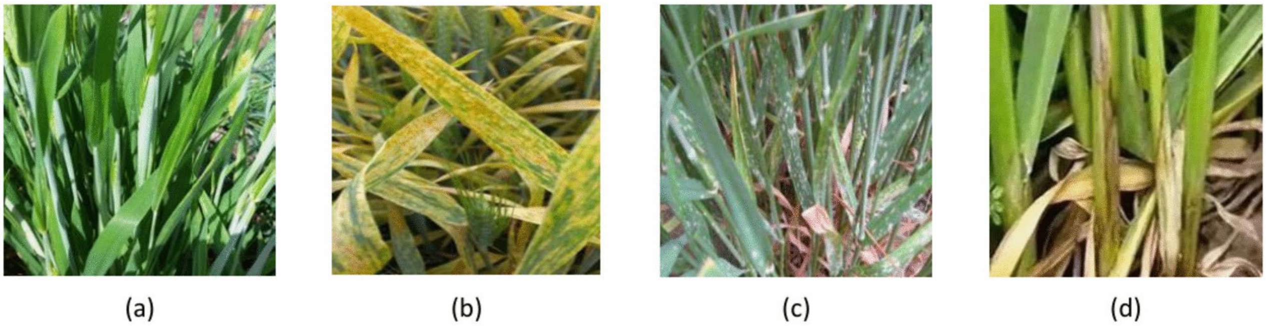

Fig. 1. Samples of wheat disease. (**a**) Healthy, (**b**) Leaf Rust, (**c**) Wheat Scab, (**d**) Wheat sharp eyespot.

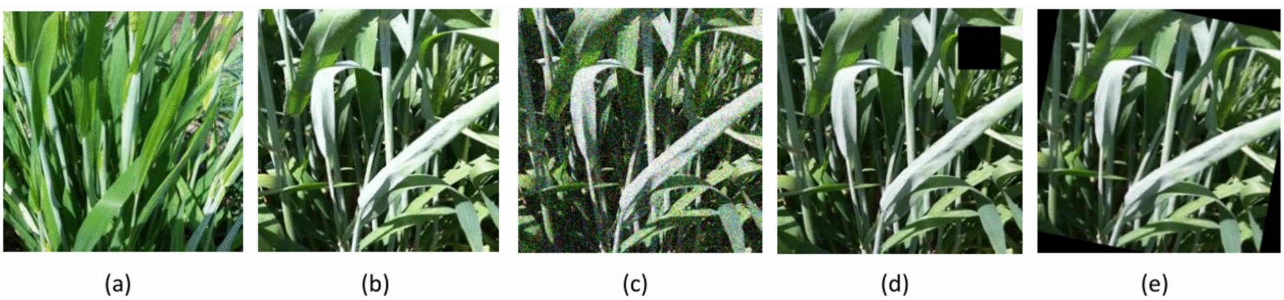

Fig. 2. Data augmentation effect. (**a**) Original images, (**b**) Random brightness, (**c**) Gaussian noise, (**d**) Random occlusion, (**e**) Rotate.

SimCLR pre-training framework

Contrastive Learning (CL) is a distinct method of self-supervised learning primarily focused on enhancing the representation of positive and negative data samples[22]. This is achieved by maximizing the similarity between relevant samples and minimizing the similarity between irrelevant samples. SimCLR is a contrastive learning framework specifically designed for visual representation pre-training. The training process of SimCLR is divided into two stages. In the first stage, the model engages in self-supervised learning using unlabeled samples, which enables it to learn similarities within the same category and differences between distinct categories. The second stage involves supervised learning with labeled samples, which refines the model's recognition of different categories. This refinement allows the model to better adapt to downstream tasks. The structural diagram of the SimCLR pre-training framework is shown in Fig. 3.

In the SimCLR pre-training framework, the images from the dataset are initially divided into multiple batches. Each batch of images undergoes two different data augmentations to produce a pair of augmented images, X_i and X_j. These images are then fed into an encoder, which computes their feature representations, hi and hj. Following this, the features pass through a non-linear fully connected layer, which transforms them into image representations Z_i and Z_j. Finally, a contrastive loss function is calculated, and the model parameters are optimized through backpropagation. The primary objective of this process is to attract the feature vectors of the same image closer to each other while pushing apart the feature vectors of different images. This approach effectively leverages the variations induced by the data augmentations to learn a robust representation of the images that can generalize well to new, unseen data.

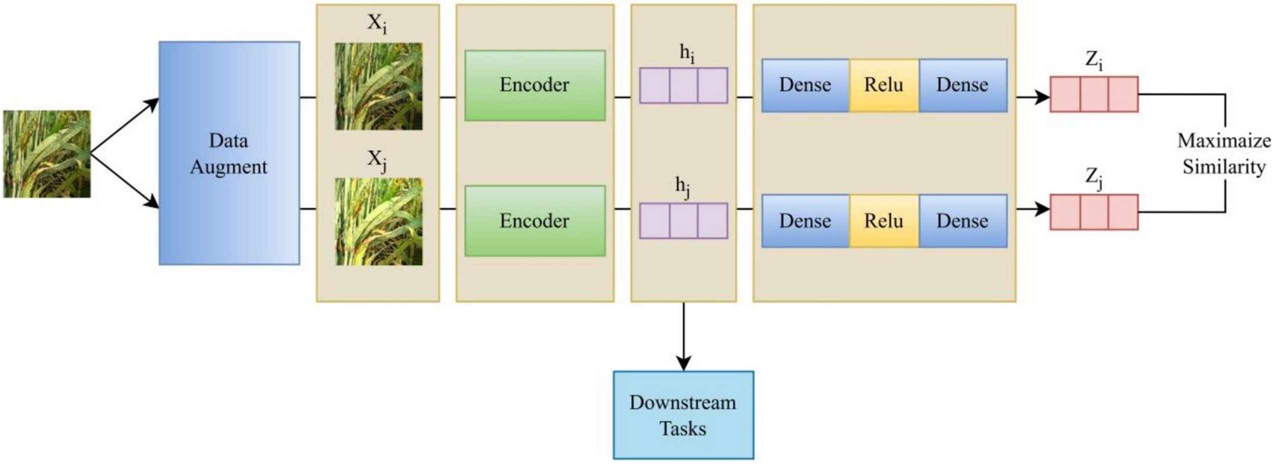

Fig. 3. SimCLR pre-training framework.

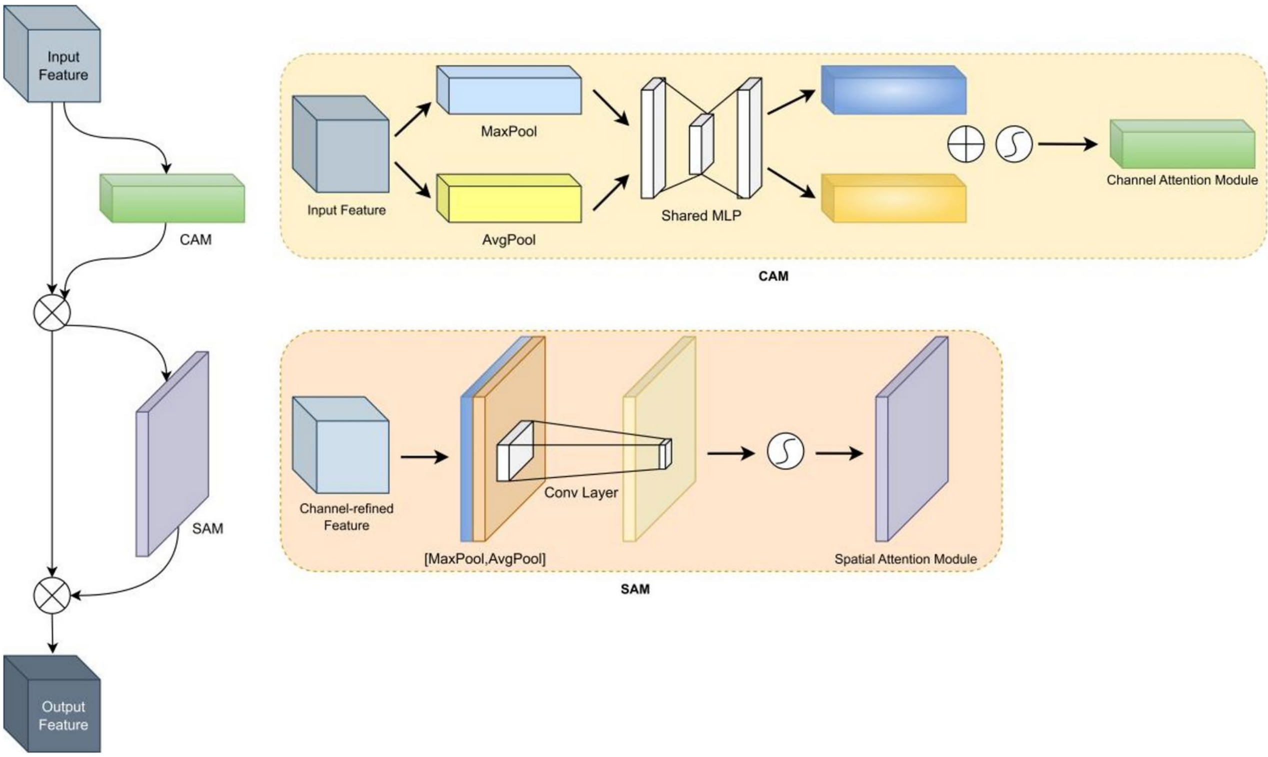

Fig. 4. CBAM structure.

Attention mechanism

Attention mechanisms are a method modeled after human visual and cognitive systems, allowing neural networks to focus limited attention on critical areas when processing input images[23]. By incorporating attention mechanisms, the model increases the computational weight assigned to regions of wheat disease and decreases the weight on areas less related to the disease. This focus allows the limited computational resources to be allocated to identifying key features, effectively filtering complex backgrounds, and enhancing the model's performance and generalization capabilities[24–27]. In this study, we introduce the Convolutional Block Attention Module (CBAM)[28] with an improved activation function to enhance the model's focus on wheat disease regions and suppress interference from complex field backgrounds. CBAM comprises a Channel Attention Mechanism (CAM) and a Spatial Attention Mechanism (SAM), which are cascaded together. As depicted in Fig. 4, CBAM extracts feature information in both channel and spatial dimensions and then merges these features through element-wise multiplication. This process results in attention-enhanced features that maximally preserve disease characteristic information and minimize information loss, thereby strengthening the model's disease detection capabilities in challenging environments.

The computational formulas for the Channel Attention Module and the Spatial Attention Module are depicted in Eqs. (1) and (2), respectively. Based on these, the overall feature processing procedure within the Convolutional Block Attention Module (CBAM) can be represented by Eqs. (3) and (4).

$$\mathrm{M_{CAM}}\left(F\right)=\sigma\left\{MLP\left[AvgPool\left(F\right)\right]+MLP\left[MaxPool\left(F\right)\right]\right\} \tag{1}$$

$$\mathrm{M_{SAM}}\left(F\prime\right)=\sigma\left\{f^{7\times7}\left(\left[AvgPool\left(F\prime\right);MaxPool\left(F\prime\right)\right]\right)\right\} \tag{2}$$

$$F\prime=M_{CAM}\left(F\right)\otimes F \tag{3}$$

$$F^{''}=M_{SAM}\left(F\prime\right)\otimes F\prime \tag{4}$$

In the equations, F, F', and F" denote the original input image, the image input processed by the Channel Attention Module, and the output from the Spatial Attention Module, respectively. The symbol σ represents the sigmoid activation function, and $\otimes$ indicates feature multiplication. Additionally, to overcome the issue of gradient vanishing caused by the inactivation of the ReLU activation function in negative regions, this study incorporates the LeakyReLU[29] activation function into the Channel Attention Module. The formula for the LeakyReLU activation function is presented as Eq. (5), and its function graph is illustrated in Fig. 5.

$$f\left(x\right)=\max\left(0.01x,x\right)=\left\{\begin{array}{c}x,x>0\\0.01x,x\leq0\end{array}\right. \tag{5}$$

Loss function

Currently, most classification networks utilize Cross Entropy Loss to compute classification loss values; however, this is only suitable for datasets where positive and negative samples are relatively balanced. When the data exhibit extreme sample imbalance, including a large number of easily classified negative samples, the use of the cross-entropy loss function can lead to significant errors in model classification. Therefore, this study introduces the Focal Loss function[30] to address potential imbalances between disease sample quantities in the wheat dataset, as well as the substantial disparities in the difficulty of classifying disease samples. The computational formula for Focal Loss is presented as Eq. (6).

$$FL\left(p_t\right)=-\left(1-p_t\right)^{\gamma}\ln\left(p_t\right) \tag{6}$$

In the above formula, p_t represents the predicted probability for the true class label, which can indicate the difficulty of classifying the sample; $(1-p_t)^{\gamma}$ acts as a weighting coefficient factor, where γ is a hyperparameter for the adjustment coefficient. This hyperparameter adjusts the significance of the loss for different samples based on the disparity between the predicted outcomes and the actual labels. The larger the value of γ, the more significant the loss for difficult-to-classify samples and less significant for easy-to-classify samples. This approach effectively addresses the issue of sample imbalance within the dataset.

SC ConvNeXt network model

This study proposes the SC-ConvNeXt network model (Fig. 6) based on the characteristics of wheat disease images. The model consists of two main components: a feature extraction network and a SimCLR pre-training framework. The feature extraction network utilizes ConvNeXt-T (the Tiny version of ConvNeXt), comprising ConvNeXt Blocks (Fig. 7) and a Downsample block, as its backbone. ConvNeXt[31] modifies the ResNet50[32] model by incorporating ResNeXt elements, replacing standard convolutions with depthwise separable convolutions to reduce computational demands. Furthermore, it adopts structural and optimization strategies from the

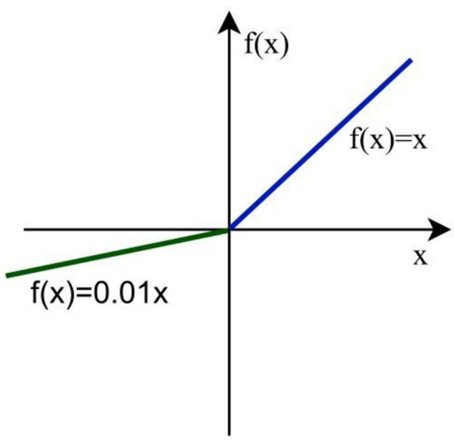

Fig. 5. LeakyReLU activation function.

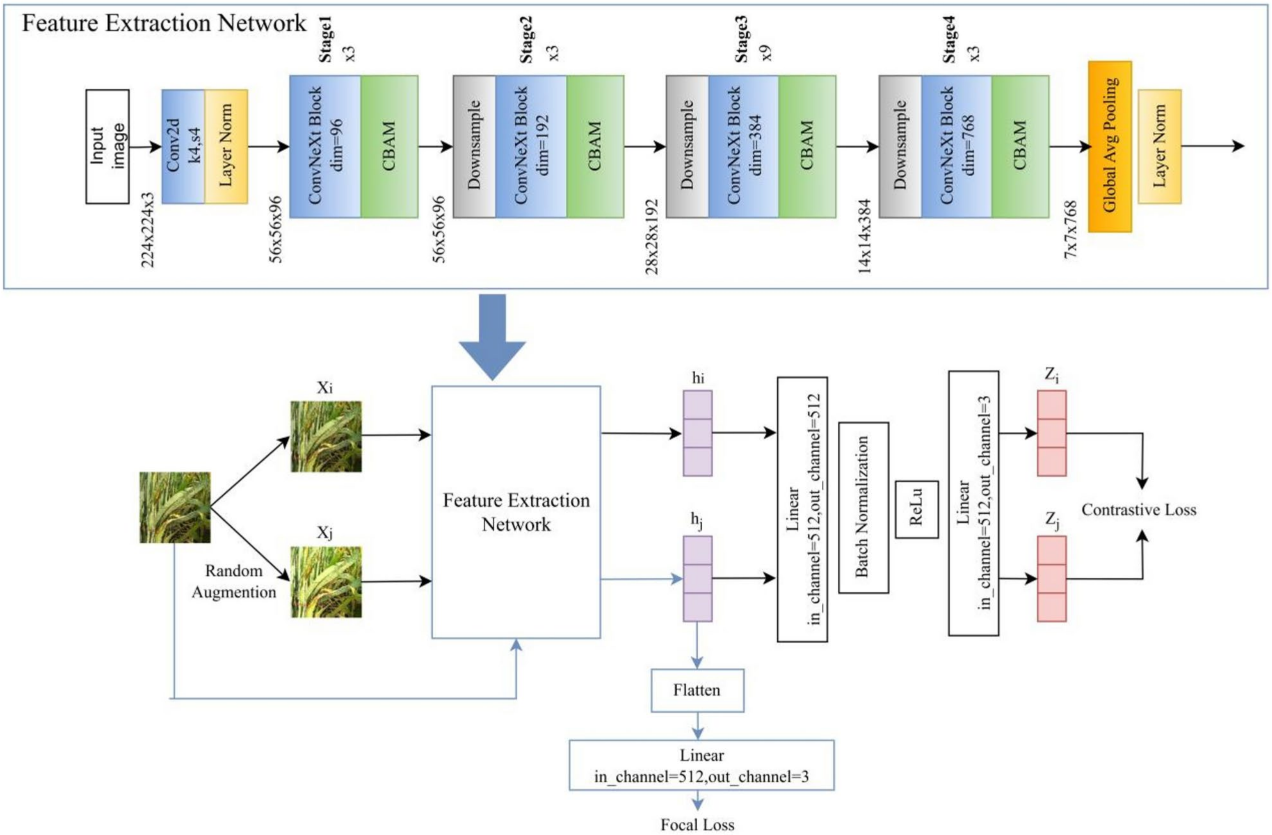

Fig. 6. SC ConvNeXt Network architecture.

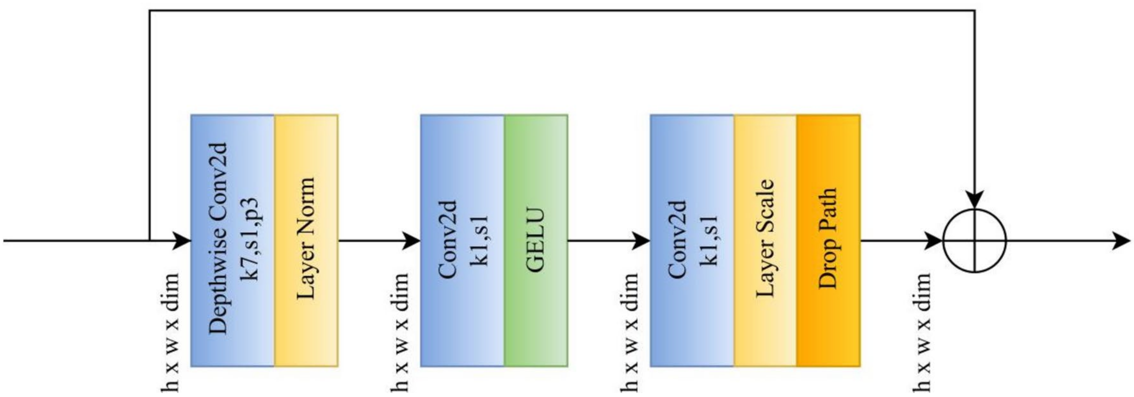

Fig. 7. ConvNeXt Block structure.

Swin Transformer[33]. On this foundation, the feature extraction network omits the final fully connected layer and embeds an improved CBAM attention module in each Stage, serving as the encoder within the SimCLR pre-training framework. The model described in this article undergoes two phases: self-supervised learning followed by supervised learning. This allows it to efficiently extract interaction information between spatial and channel dimensions while using fewer labeled disease samples, thereby achieving accurate identification of wheat diseases.

In the self-supervised learning phase, the process begins by applying two distinct random data augmentations to a set of input images that lack disease category labels. The augmented images are then subjected to feature extraction through an encoder, generating two sets of image representations, X_i and X_j. Subsequently, these representations pass through a fully connected layer with an input size of 512 and an output of 512, followed by a batch normalization layer, a ReLU activation function, and another fully connected layer with an input size of 512 and an output of 4. The resulting feature representations, Z_i and Z_j, are then input into a contrastive loss function to compute the loss value. A smaller contrastive loss value indicates a higher probability that the two images belong to the same class, whereas a larger loss suggests they might originate from different samples. The model updates the gradients by utilizing the loss value, specifically by maximizing the similarity between Z_i and Z_j, which are different feature representations of the same sample, through the contrastive loss function. This maximization completes the pre-training of the network in the first stage. The contrastive loss function is expressed as Eq. (7).

$$l_{i,j} = -\log \frac{\exp\left(sim\left(z_i, z_j\right)/\tau\right)}{\sum l_{[k \neq i]} \exp\left(sim\left(z_i, z_j\right)/\tau\right)} \tag{7}$$

In this formula, Z_i and Z_j represent feature vectors transformed by a nonlinear function; $l_{[k \neq i]} \in \{0, 1\}$ is an indicator function that takes a value of 1 when $k \neq i$ and 0 otherwise; τ represents a constant; $sim()$ is a function calculating cosine similarity, formally defined as $sim\left(u, v\right) = \frac{u^T v}{\|u\| \|v\|}$.

In the second phase, supervised learning, as illustrated by the blue line in Fig. 7, the encoder retains the parameters obtained during the self-supervised learning phase. It extracts features from the input images, producing feature vectors that are subsequently flattened and fed into a linear fully connected layer for classification. The loss is calculated using the focal loss function. This phase involves training with labeled samples to fine-tune the network model.

Evaluation metrics

To more directly and objectively evaluate the performance of the network model proposed in this study, the experiment employs four key metrics: Accuracy, Precision, Recall, and F1-score. Accuracy represents the proportion of correct predictions among the total number of cases, but in situations with imbalanced samples, relying solely on accuracy may not adequately reflect the quality of the model. Precision indicates the ratio of correctly predicted positive instances among all positive predictions made. Recall denotes the proportion of correctly predicted positive instances among all actual positive instances. The formulas for calculating Accuracy, Precision, Recall, and F1-score are defined as follows:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{8}$$

$$Precision = \frac{TP}{TP + FP} \tag{9}$$

$$Recall = \frac{TP}{TP + FN} \tag{10}$$

$$F1 - score = 2 \times \frac{Precision \times Recall}{Precision + Recal} \tag{11}$$

The terms TP (True Positive), FP (False Positive), TN (True Negative), and FN (False Negative) represent the following outcomes in prediction: TP indicates a correct prediction of a positive sample, FP denotes an incorrect prediction of a positive sample, TN refers to a correct prediction of a negative sample, and FN signifies an incorrect prediction of a negative sample.

A confusion matrix provides a straightforward visual representation of the relationship between predicted outcomes and actual class labels, thereby illustrating the effectiveness of a model's classification ability. In a confusion matrix, each column represents predicted categories, while each row represents actual categories of the samples. Data concentrated along the diagonal, where the values on the diagonal are higher, indicates a better classification performance by the model. Therefore, in this experiment, the confusion matrix is used to observe the model's classification performance and to identify disease categories that are prone to confusion. This tool allows for a clear and quantifiable assessment of both the strengths and weaknesses of the classification system in differentiating between various types of wheat diseases.

Results and discussion

This experiment is based on the PaddlePaddle 2.3.2 deep learning framework and uses Python 3.7.4. The computational resources include a 4-core CPU and a Tesla V100 GPU with 32GB of VRAM.

The settings for the SC-ConvNeXt model parameters are as follows: both the self-supervised and supervised learning phases are set to run for 100 training iterations. Batch sizes are set at 64 for unsupervised learning and 32 for supervised learning, with the test set also having a batch size of 32. The self-supervised learning phase employs a contrastive loss function, while the supervised learning phase utilizes the Focal Loss function. The Adam optimizer is used with an initial learning rate set at 0.00001[34].

To validate the performance of the SC-ConvNeXt model, this study includes two sets of ablation experiments: a comparison of the effects before and after the addition of the SimCLR pre-training framework, and a comparison of recognition performance before and after the incorporation of an attention mechanism. Additionally, comparative experiments are conducted against other mainstream classical network models to evaluate performance.

Analysis of the experimental results of adding the SimCLR pre-training framework

To verify the positive impact of incorporating the SimCLR pre-training framework on the classification and recognition of wheat diseases, this study conducted experiments with and without SimCLR on the improved

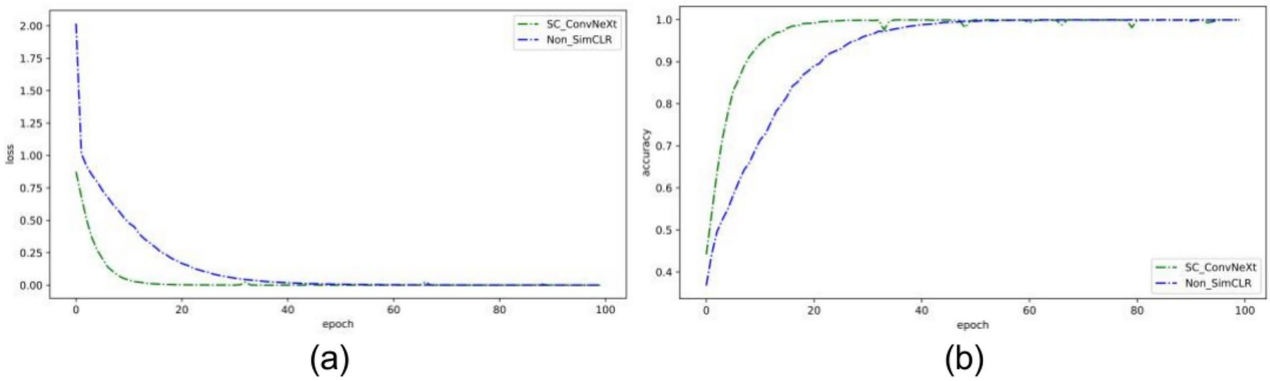

Fig. 8. Append the curves depicting changes in loss values and accuracy rates before and after the addition of the SimCLR pre-training framework. (**a**) Include the loss value variation curves before and after the incorporation of the SimCLR framework. (**b**) Include the accuracy variation curves before and after the implementation of the SimCLR framework.

Models	Accuracy(%)	Precision(%)	Recall(%)	F1-score(%)
SC-ConvNeXt	88.05	88.18	88.05	88.05
Non-SimCLR	86.98	87.09	86.98	87.01

Table 2. Comparison of experimental results before and after the addition of the SimCLR pre-training framework.

network model. During the training process (all supervised learning), the changes in loss and accuracy before and after adding the SimCLR pre-training framework are shown in Fig. 10. According to Fig. 8, the network model integrated with SimCLR exhibited more than a 1% increase in accuracy and a reduction in loss of over 1.2 compared to the model without SimCLR. Furthermore, the loss and accuracy curves of the SimCLR-integrated network model were smoother and converged more rapidly. These results demonstrate that adding the SimCLR pre-training framework can enhance model convergence, thereby improving the training speed and effectiveness of the model. The reason is that, after the first round of self-supervised learning with the addition of the SimCLR pre-training framework, the feature extraction network has undergone preliminary learning of wheat disease characteristics and has become familiar with the distinct features of each disease. As a result, the model demonstrates a higher recognition rate for diseases within the same category and achieves faster learning in subsequent supervised learning phases.

Table 2 presents a comparison of the experimental results of the network model before and after incorporating the SimCLR pre-training framework on the test set. As shown by the evaluation metrics in Table 2, the addition of the SimCLR pre-training framework has led to significant improvements in the model's recognition accuracy, precision, recall, and F1-score, with increases exceeding 1%. This demonstrates that incorporating the SimCLR pre-training framework into the feature extraction network proposed in this study can effectively enhance the performance of the wheat disease recognition model without requiring additional labeled training samples. Compared to traditional network model training, this approach optimizes parameters during an initial unsupervised learning phase before supervised learning, further improving the performance of the wheat disease recognition network.

Comparison of recognition performance before and after the use of the improved CBAM attention mechanism

To verify the effectiveness of the improved CBAM attention mechanism proposed in this study for enhancing the feature extraction capability of disease characteristics, an ablation study was conducted. Under the condition that other variables remained constant, the only change made was the addition or non-addition of the attention mechanism, to test the model's performance in identifying wheat diseases. The comparison of loss values and accuracy curves before and after the addition of the attention mechanism is shown in Fig. 9, and the various evaluation metrics are presented in Table 3. After the incorporation of the attention mechanism, the network model demonstrated a significant improvement in training convergence speed and recognition performance, with a more stable convergence process. According to the analysis of experimental data, the feature extraction network with added attention mechanism can more efficiently allocate model weights, focus the network's attention on the affected areas of the wheat, and enable the model to precisely capture the spatial and channel features of wheat diseases against complex field backgrounds.

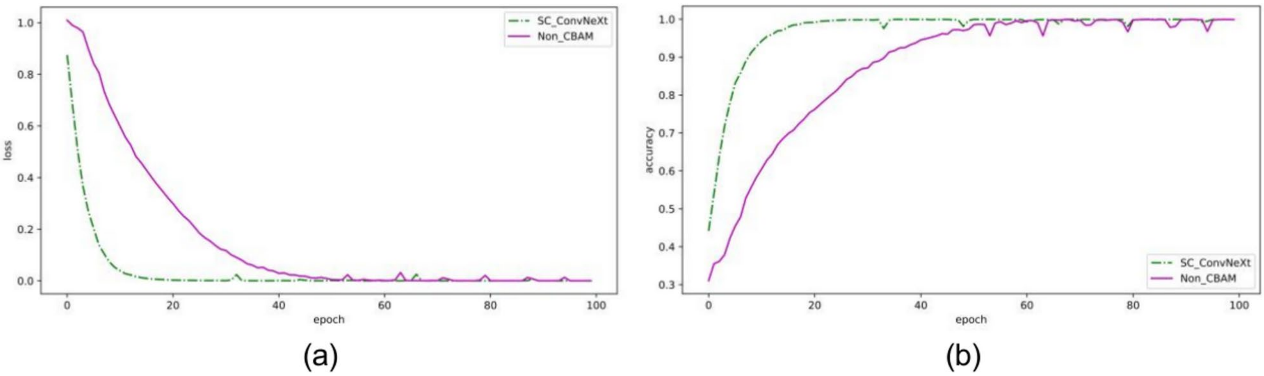

Fig. 9. Curves illustrating the changes in loss values and accuracy before and after the use of the improved CBAM attention mechanism. (**a**) Loss value variation curves before and after the implementation of the attention mechanism. (**b**) Accuracy variation curves before and after the implementation of the attention mechanism.

Models	Accuracy(%)	Precision(%)	Recall(%)	F1-score(%)
SC-ConvNeXt	88.05	88.18	88.05	88.05
Non-CBAM	84.47	84.67	84.47	84.49

Table 3. Comparison of experimental results before and after the application of the improved CBAM attention mechanism.

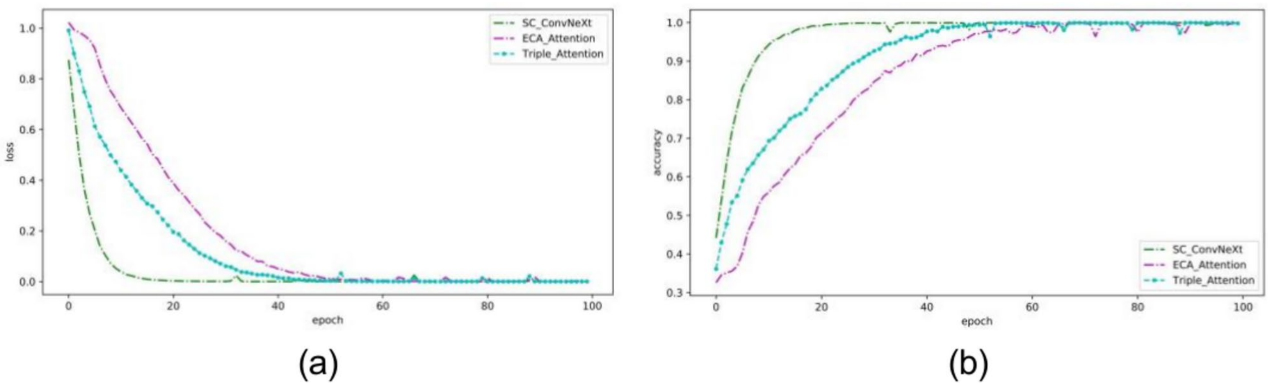

Fig. 10. Add the curve of loss values and accuracy changes for different attention mechanisms. (**a**) Add the change curve of the loss value of different attention mechanisms. (**b**) Add the curve of accuracy changes for different attention mechanisms.

Models	Accuracy(%)	Precision(%)	Recall(%)	F1-score(%)
SC-ConvNeXt	88.05	88.18	88.05	88.05
ECA Attention	85.75	85.96	85.75	85.81
Triple Attention	87.43	87.46	87.43	87.43

Table 4. Comparison of the effects of experiments that add different attention mechanisms.

Comparison of recognition effectiveness using the improved CBAM attention mechanism with other attention mechanisms

To demonstrate the superiority of the improved CBAM attention mechanism, the modified CBAM attention module, the ECA attention model, and the three-branch attention module were added to the feature extraction network at the same position. The loss values and accuracy curves of the models with different attention mechanisms during training are shown in Fig. 10, with evaluation metrics provided in Table 4. As observed in

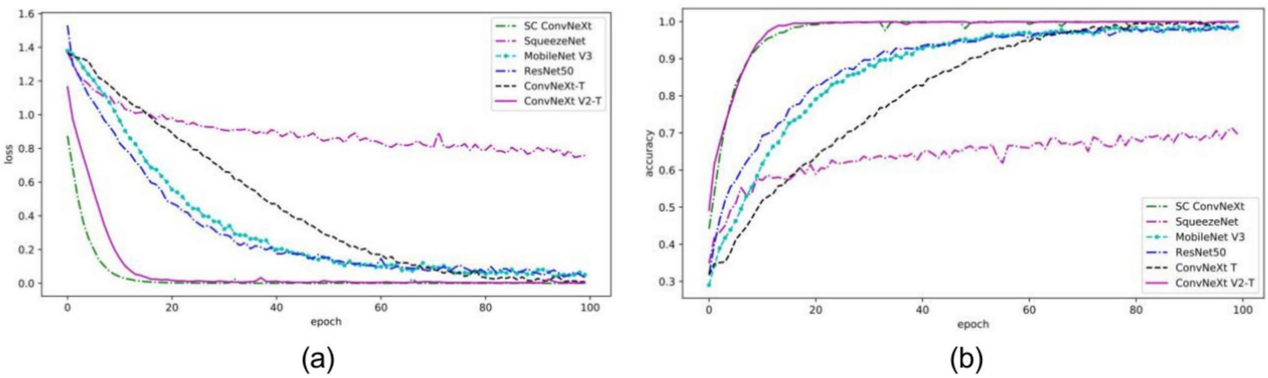

Fig. 11. Comparison of the variation curves for loss values and accuracy across different models. (**a**) Loss value variation curves for different models. (**b**) Accuracy variation curves for different models.

Models	Accuracy(%)	Precision(%)	Recall(%)	F1-score(%)	Parameter quantity(MB)
SC-ConvNeXt	88.05	88.18	88.05	88.05	216.37MB
SqueezeNet	71.38	71.31	71.38	71.03	2.81MB
MobileNetV3	74.82	74.75	74.82	74.77	16.14MB
ResNet50	77.39	77.75	77.39	77.28	89.91MB
ConvNeXt-T	84.22	84.34	84.22	84.26	106.14MB
ConvNeXt V2-T	85.86	85.90	85.86	85.87	106.16MB

Table 5. Comparison of experimental results among various models.

Fig. 10, the model with the improved CBAM attention module exhibits a significantly faster convergence speed than the other two attention mechanisms. The data in Table 4 indicate that the networks with the ECA attention mechanism and the three-branch attention mechanism are inferior to the current model in all evaluation metrics, particularly in recognition accuracy, where they trail by approximately 2.3% and 0.6%, respectively. These experimental results suggest that the improved CBAM attention mechanism offers the best enhancement for model performance.

Comparison of recognition performance across different network models and confidence test

To validate the classification effectiveness of the proposed SC-ConvNeXt model on wheat diseases, comparative experiments were conducted using the same dataset with SC-ConvNeXt, SqueezeNet, MobileNetV3, ResNet50, ConvNeXt-T and ConvNeXt V2-T models under identical experimental conditions. The comparison curves of loss values and accuracy during the training phase for different models are illustrated in Fig. 11. The classification results of the different models are presented through confusion matrices, with comparisons of these matrices also shown in Fig. 11. The comparative experimental results of each model are displayed in Table 5.

As illustrated in Fig. 11, from the beginning to the end of training, the SC-ConvNeXt model consistently outperforms the other four models in terms of accuracy and loss values, demonstrating faster convergence as well. As shown in Fig. 12, compared to other models, the SC-ConvNeXt model's confusion matrix displays darker colors along its diagonal, indicating that most classification results are concentrated along the diagonal with a significantly higher number of correct identifications for various diseases, particularly for easily confused diseases such as Leaf Rust (label 1) and Wheat Sharp Eyespot (label 3). The proposed model exhibits a notable advantage in recognizing these diseases. According to Table 5, the SC-ConvNeXt model achieves an accuracy of 88.05%, which is a 3.83% improvement over the ConvNeXt-T model at 84.22%, and 16.67% higher than the SqueezeNet model. Moreover, it has the highest precision, recall, and F1-score among the five models compared. However, due to varying degrees of disease severity and the similarity between diseased and healthy wheat, coupled with natural environmental factors like lighting and weeds affecting the dataset captured in natural field conditions, some classification errors are inevitable. Nevertheless, the SC-ConvNeXt model maintains a relatively high level of classification performance.Furthermore, as this model comprises two main components—a SimCLR pre-training framework and a feature extraction network—the parameter count is approximately 7.5 times that of the SqueezeNet model and twice that of the ConvNeXt-T model. However, with the ample availability of modern computer storage resources, the SC-ConvNeXt network model achieves improved recognition accuracy and other benefits with a moderate increase in parameters, making this outcome satisfactory.

Furthermore, the SC-ConvNeXt network model is designed to maintain consistent performance in diverse and challenging environments. This study employed a repeated measures design, with 10 experiments conducted on the model. The accuracy of each experiment was recorded, and the mean and confidence interval were calculated. The results are shown in Table 6. As demonstrated by the data in the table, the results of the 10 experiments show minimal variation, consistently averaging around 88%, with a 95% confidence interval of

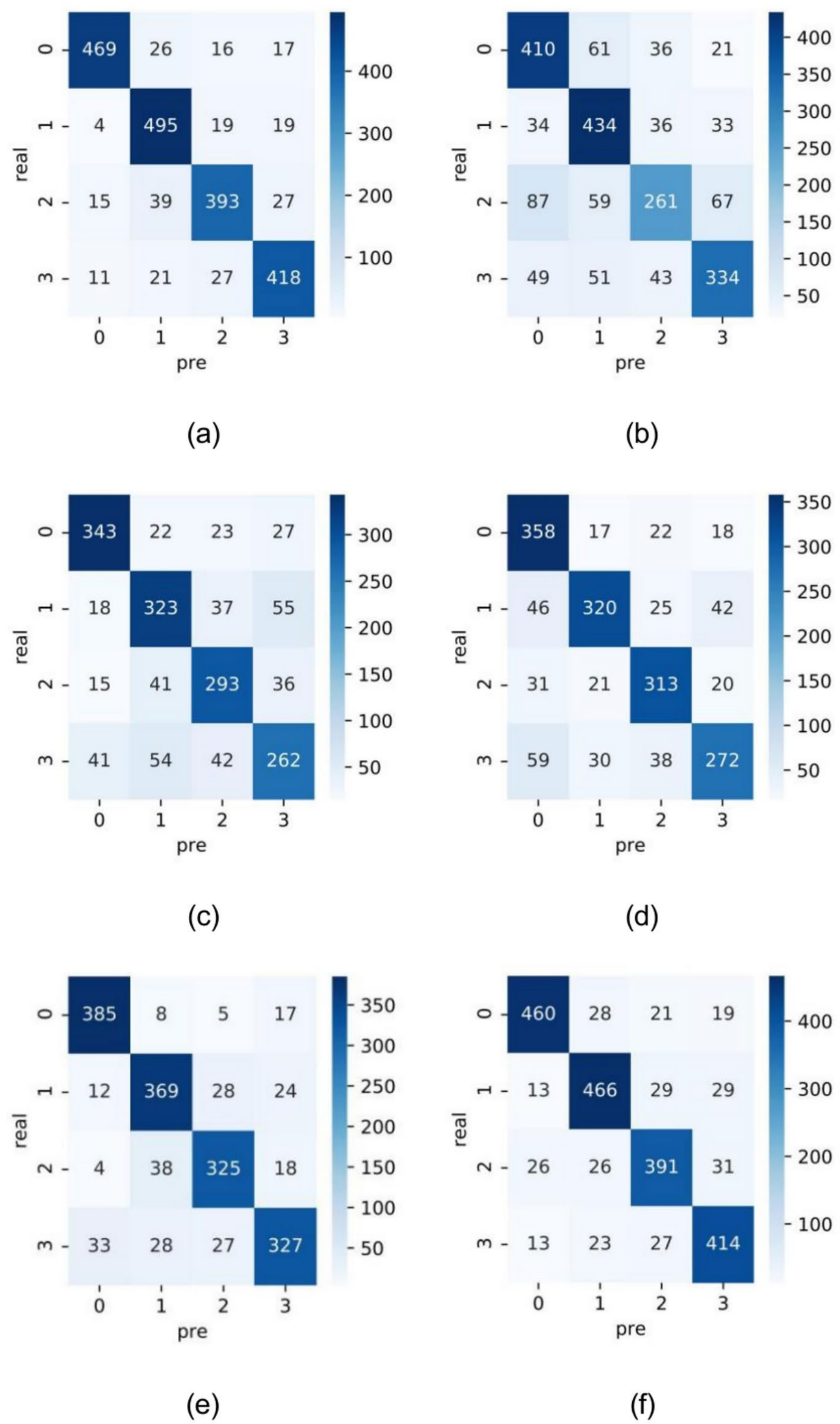

Fig. 12. Comparison of confusion matrices across different models. (**a**) SC-ConvNeXt, (**b**) SqueezeNet, (**c**) MobileNetV3, (**d**) ResNet50, (**e**) ConvNeXt-T.

Model	Accuracy(%)					Average accuracy(%)	95% Confidence interval
SC-ConvNeXt	88.05	87.94	88.03	88.01	87.85	88.00	(0.8793,0.8807)
	88.04	88.10	88.06	87.87	88.05		

Table 6. Confidence test data.

(0.8793, 0.8807). This indicates that the SC-ConvNeXt network exhibits strong stability, supporting its ability to perform reliably in future recognition tasks.

Outlook

The experimental data show that the SC-ConvNeXt model proposed in this paper not only performs excellently in recognizing wheat diseases but also reduces reliance on labeled data in the training process. When deployed on a specific device or platform and applied to disease recognition scenarios in real agricultural environments, the model can be trained with only a limited amount of labeled data and field-collected images. Following this, the model is ready to perform disease recognition. This advantage optimizes data utilization for crop disease recognition across large areas, effectively reducing model training costs. Furthermore, the impact of external factors, such as climate and light variations, on the disease recognition process should be considered. To address this, the experiment employs data augmentation techniques to simulate diverse environmental conditions, enhancing the model's adaptability and overall performance.

It should be noted that this method has certain limitations. The model has a relatively large number of parameters and follows a two-stage training process, resulting in longer training times for the SC-ConvNeXt model. Additionally, the computational capacity of field equipment is essential for training. These limitations warrant further investigation to explore potential solutions for model compression and parameter updates. Future research could enhance the model by integrating more advanced image classification and self-supervised learning techniques. For example, optimizing the feature extraction network in the ConvNeXt V2 model could improve both accuracy and efficiency. Additionally, replacing the self-supervised learning frameworks with more efficient alternatives, such as BYOL[35] and Masked Autoencoders (MAE)[36], could further enhance performance.

Despite these limitations, the current SC-ConvNeXt model demonstrates significant potential and value for extensive deployment in diverse agricultural applications.

Conclusion

This paper combines deep learning with crop disease prevention and control, proposing the SC-ConvNeXt wheat disease identification model. This network model uses ConvNeXt-T as the feature extraction network and incorporates an improved CBAM attention mechanism to enhance the model's focus on disease features from both spatial and channel dimensions, thereby reducing interference from complex background factors. It integrates the SimCLR pre-training framework to enhance the network model's training convergence speed and classification performance without the need for additional labeled data. Furthermore, a focal loss function is introduced to balance the classification difficulty among samples. The results from comparative and ablation experiments demonstrate that the proposed wheat disease identification model outperforms other classic classification networks in various evaluation metrics, achieving an average classification accuracy of 88.05% on the test set, which is higher than that of comparative models under the same experimental conditions. The introduction of this model promotes the integration of deep learning technology with agricultural production, making digital disease prevention and control for wheat more aligned with real-world needs.

Data availability

Since the 'Smart Agriculture' platform of Jilin Agricultural Science and Technology College is not open to the public, the wheat dataset involved in this study cannot be made publicly available. Should there be reasonable requests, please contact the corresponding author, You Tang, for access.

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T Dong wrote the main manuscript text , T Dong and Xiao Ma did the experiments, B Huang prepared Fig. 3, W Zhong prepared Fig. 4, Q Han prepared Fig. 5, Y Tang prepared Figs. 6–7.All authors reviewed the manuscript.

Competing interests
The authors declare no competing interests.

Additional information
Correspondence and requests for materials should be addressed to Y.T.

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